Stephen Nyaranga

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EDUCATION

African Institute for Mathematical Science| Research and Innovation Center (AIMS-RIC) **Oct 2025-To date**
Kigali, Rwanda/ Imperial College London

- *PhD Student, Climate Science*
- ***Research Topic:*** *Multi-Sensor Deep Learning Framework for Flood Depth Estimation in Mountainous Cities: Integrating SAR Time-Series, Optical EO, and Terrain Metrics in Musanze & Rubavu, Rwanda.*

International Center for Theoretical Physics-East Africa **Jan 2023- Dec 2024**
Institute for Fundamental Research, ICTP-EAIFR- Kigali, Rwanda

- MSc. Geophysics

African Institute for Mathematical Science (AIMS)-SENEGAL – Mbour, Senegal **Oct 2021- July 2022**

- MSc. in Mathematical Sciences- Data Science
- Graduated with Distinction

Karatina University – Nyeri, Kenya **Aug 2016-March 2021**

- BSc. Education Science (Physics/Mathematics)
- Graduated with First Class Honors

PROFESSIONAL EXPERIENCE

VATEL RWANDA –Kigali, Rwanda **July 2024- Oct 2025**
Head of IT Department — Full Time

- Contributed to academic quality assurance and digital transformation of teaching.
- Designing and Developing the Learning Management System and Data Controller.

VATEL RWANDA –Kigali, Rwanda **May 2025- Oct 2025**
Assistant Lecturer — Part Time

- Prepared lessons and teaching materials for ICDL for students and staff.
- Facilitated AI integration in curriculum and staff development.

VATEL RWANDA –Kigali, Rwanda **July 2023- July 2024**
IT Associate Intern- UNDER MASTERCARD FOUNDATION GRADUATE ENRICHMENT PROGRAM

- Prepared lessons and teaching materials for IT students and staff.
- Performing IT roles.

TEACHER KATHIANI BOY'S HIGH SCHOOL – Machakos, Kenya **June 2021 – Aug 2021**
TEACHER NTANKI SECONDARY SCHOOL– Meru, Kenya **Jan 2021 - April 2021**

- Prepared lessons and teaching materials, besides assessing students' progress in the following subjects: Mathematics and Physics.
- Working as a Physics/Mathematics teacher helped refine my negotiator and motivator skills.

SKILLS AND INTERESTS

- Fluent in English, Swahili, and Luhya
- Advanced skills in Excel, Online Collaboration & PowerPoint
- Python Programming, C/C++ and R-Language, Linux Command, LaTeX, MATLAB, Fortran, and Microsoft Office.
- Excellent Communicator, Great interpersonal skills, Complex problem solver, Innovator, and Service Oriented

Objective

- A diligent young professional looking to apply the knowledge acquired at school to deliver value to the organization and personal career growth and development.

CONFERENCE, WORKSHOPS, TRAINING

1. **Ministry of Education, Rwanda Statistical Data Preparation and Analysis Workshop: Sept 30th–4th Oct 2024**
   Completed courses on data controllers for tertiary-level education.
2. **Jobberman's Finishing School Program** **May 8th – 19th 2023**
   Completed courses in Work Ethics and Professionalism, Business Writing and Communication, Professional Attitude & Growth Mindset, and Personal Effectiveness & Time Management.
3. **International Computer Driving License (ICDL) Certification** **July 2023 – Nov 2023**
   ICDL trainer at Vatel Rwanda for Students and Staff, i.e., Cybersecurity, AI, Microsoft Office, Online Collaboration, Big Data, ICT in Education, Computer and Online Essentials.

4. **African School of Fundamental Physics and Applications (ASP)** **Nov 2022 – Dec 2022**

Renewable energies, High Energy Physics, Astronomy, and Cosmology.

5. **7th African School on Electronic Structure Methods and Applications - ASESMA 2023** **12th – 23rd Jun 2023**
6. **ICTP-EAIFR-IUGG Workshop on Computational Geodynamics: Towards Building a New Expertise Across Africa** **2-7 Jul 2023**

Awards, Scholarships, and Honors.

- ICTP-EAIFR (UNIVERSITY OF RWANDA)- Best winner 3-Minute Thesis Award at the ICTP-EAIFR PhD and master's Thesis Competition. **July 22, 2024**
- Mastercard Foundation Scholarship; MSc at AIMS-Senegal. **Oct 2021–Jul 2022**

Highlight of Some Graduate Courses.

- Climate Dynamics, Machine Learning, Classical Mechanics, Atmospheric Physics, Ordinary differential equations, Statistical Mechanics, Continuum Mechanics, Quantum Mechanics I & II, Artificial Intelligence, Classical Mechanics, Mathematical Methods of Physics, Physics of Volcano, Earthquake Mechanics and Tectonophysics, Advanced Electromagnetism, Theoretical and Observational Seismology, GPS Geodesy and InSAR, Fundamentals of Solid Earth, and Partial Differential Equations.

Membership.

- Teachers Service Commission (TSC)-Kenya.
- African School of Fundamental Physics and Applications (ASP).
- African Strategy for Fundamental and Applied Physics (ASFAP).